LECTURE 21

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Show that limit for following does not exist:

1) $f(x,y) = \dfrac{y-x}{y+x} \times \dfrac{1+x}{1+y}$ 2) $f(x,y) = \dfrac{2xy^2}{x^3+3y^3}$

Continuity: $\lim\limits_{(x,y)\to(a,b)} f(x,y) = f(a,b)$

1) $f(x,y) = \begin{cases} \dfrac{xy}{\sqrt{x^2+y^2}} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$ at $(0,0)$

$f(x,y) = \dfrac{r\cos\theta \; r\sin\theta}{r} = r\sin\theta\cos\theta = \dfrac{r}{2}\sin 2\theta$

$\lim\limits_{(r,\theta)\to(0,0)} = 0$

$\lim\limits_{(x,y)\to(0,0)} f(x,y) = 0 = f(0,0)$

2) $f(x,y) = \begin{cases} x^2+2y & (x,y) \neq (1,2) \\ 3 & (x,y) = (1,2) \end{cases}$ at $(1,2)$

Partial derivatives:

$\dfrac{\partial f}{\partial x}(a,b) = \lim\limits_{h\to 0} \dfrac{f(a+h,b) - f(a,b)}{h}$

$\dfrac{\partial f}{\partial y}(a,b) = \lim\limits_{k\to 0} \dfrac{f(a,b+k) - f(a,b)}{k}$

1) $f(x,y) = \begin{cases} \dfrac{xy}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$

$\dfrac{\partial f}{\partial x}(0,0) = \lim\limits_{h\to 0} \dfrac{f(h,0) - f(0,0)}{h} = \lim \dfrac{0-0}{h} = 0$

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h\to 0} \frac{\quad}{h} = \lim_{h\to 0} \frac{0-0}{h} = 0$$

$$\frac{\partial f}{\partial y}(0,0) = \lim_{k\to 0} \frac{f(0,k)-f(0,0)}{k} = \lim_{k\to 0} \frac{0-0}{k} = 0$$

<u>Note</u>: f not conti but partial derivative at $(0,0)$ exist.

2) $f(x,y) = |x| + |y|$ $\quad (x,y) \neq (0,0)$

0 $\quad (x,y) = (0,0)$ $\quad \lim_{x\to 0} \frac{|x|}{x}$?

$$\frac{\partial f}{\partial x}(0,0) \to f_x = \lim_{h\to 0} \frac{|h| + 0 - 0}{h} = \lim_{h\to 0} \frac{|h|}{h} \text{ does not exist}$$

$$\frac{\partial f}{\partial y}(0,0) \to f_y = \lim_{k\to 0} \frac{f(0,k) - f(0,0)}{k} = \lim_{k\to 0} \frac{|k|}{k} \text{ does not exist}$$

H.W. $f(x,y) = \dfrac{x^2 - y^2}{x^2 + y^2}$ $\quad (x,y) \neq (0,0)$

0 $\quad (x,y) = (0,0)$ $\quad \lim_{x\to 0} \frac{1}{x}$

<u>Ans</u>: $f_x(0,0) = \frac{1}{h}$ $\quad f_y(0,0) = \frac{-1}{k}$ $\quad$ does not exist.

4) $f(x,y) = x^4 y^3$

$$f_x = \frac{\partial f}{\partial x} = y^3 \frac{\partial f}{\partial x} x^4 = y^3 \cdot 4x^3 = 4x^3 y^3$$

$$f_y = \frac{\partial f}{\partial y} = x^4 \frac{\partial f}{\partial y} y^3 = x^4 \cdot 3y^2 = 3x^4 y^2$$

H.W. Find f_x, f_y for $f(x,y) = \dfrac{xy}{x^2+y^2}$